Additional Recurrence Relations Problems

Filename: `No File Submission`

Instructions

This worksheet contains additional practice problems on solving recurrence relations. Try to complete each problem on your own before looking at the solutions. The solutions will be provided separately, so make sure to give each problem an honest attempt first. Checking the answer too early defeats the purpose of the practice. As you work, time yourself and do not spend more than 10 minutes on any single problem.

TODO 1

Given the following function definition of `factorial`, write the recurrence relation for its running time $T(n)$. Then, use the iteration technique to determine its Big-Oh running time.

```
int factorial(int n) {
    if( n == 0 )
        return 1;
    else
        return n * factorial(n - 1);
}
```

.

TODO 2

Use the iteration technique to determine what the recurrence looks like at the kth step. **Hint:** You will need to use a summation.

$$T(n) = 5T(n-1) + n^3$$

TODO 3

Use the iteration technique to determine the Big-Oh run-time of the following recurrence relation.

$$T\left(n\right) = 2T\left(\frac{n}{2}\right) + 1 \qquad\qquad T\left(1\right) = 1$$

TODO 4 (Challenging)

Use the iteration technique to determine the Big-Oh run-time of the following recurrence relation.

$$T(n) = 3T\left(\frac{n}{4}\right) + n^2 \qquad\qquad T(1) = 1$$

TODO 5

Use the iteration technique to determine the Big-Oh run-time of the following recurrence relation.

$$T(n) = nT(n-1) \qquad\qquad T(0) = 1$$

TODO 6 (Challenging)

Use the iteration technique to determine the Big-Oh run-time of the following recurrence relation. You may assume that the input value of n is always an even number.

$$T(n) = T(n-2) + n^2 \qquad\qquad T(0) = 0$$